



University of Wisconsin
**SCHOOL OF MEDICINE
AND PUBLIC HEALTH**

BMI/STAT-768

Statistical Methods for
Medical Image Analysis and
High Dimensional Data

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<https://github.com/laplcebeltrami/BMI768>

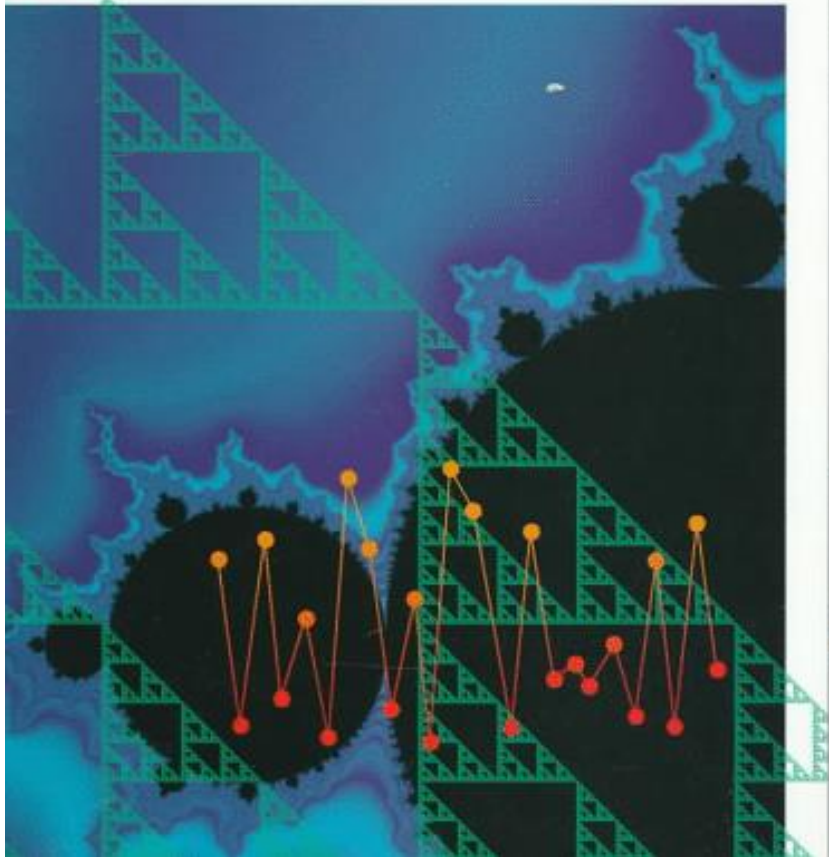
Data Complexity

Unit Objectives

1. Understand **fractal dimension**
2. Understand how to relate data **scale** and **data dimension**

Reference

Fractals and Chaos
Simplified for the Life Sciences
Larry S. Liebovitch



Benoit B. Mandelbrot (1960). Fractal geometry of nature

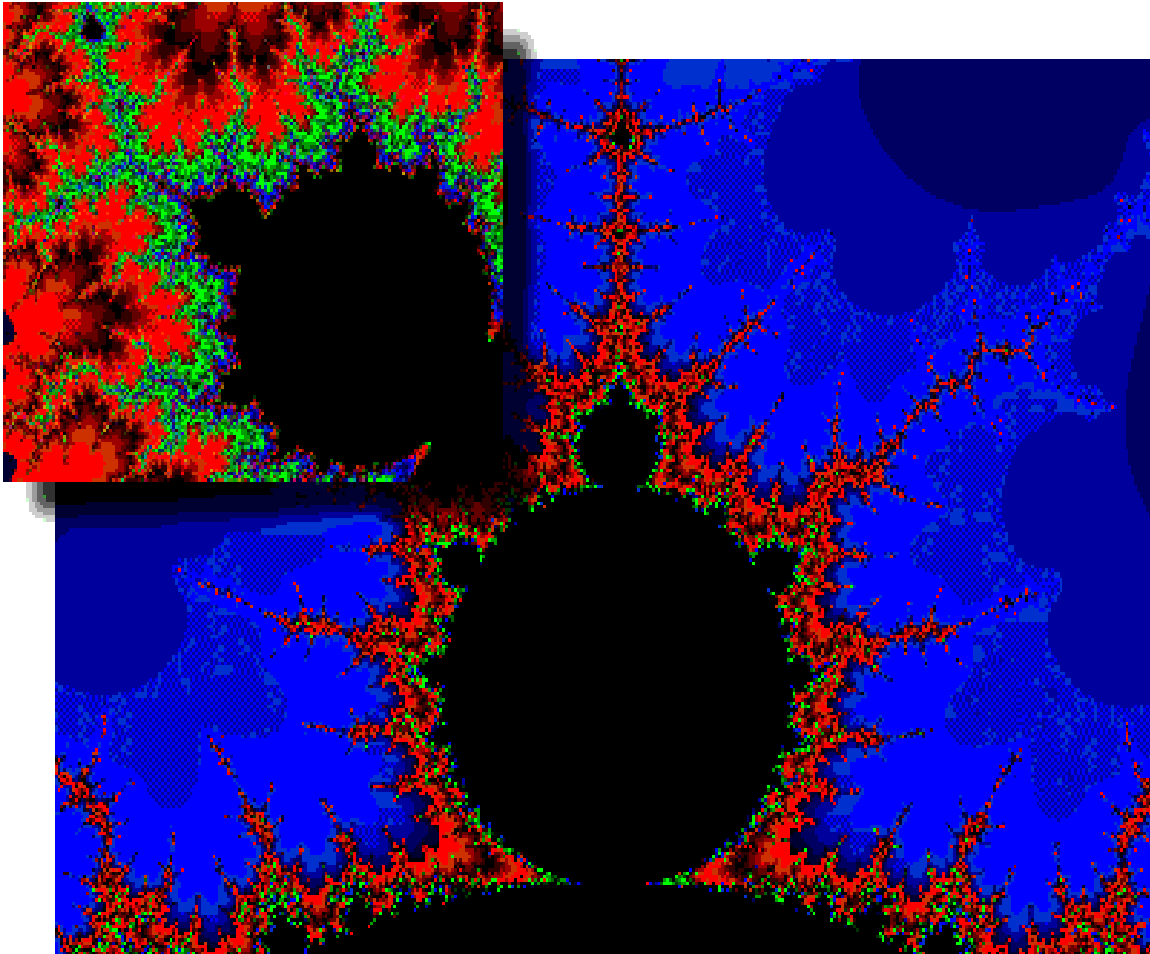
Larry S. Liebovitch 1998
Fractals and Chaos and
Simplified for the Life Sciences
Oxford Univ. Press

Required reading for data complexity

El-Yaagoubi et al. 2026
[bioRxiv:2026.05.06.723123](https://doi.org/10.1101/2026.05.06.723123)

Protein folding

Fractals generated by complex numbers



Fractals can be generated by recursive formula in a complex plane.

For instance, the following formula will generate the left fractal:

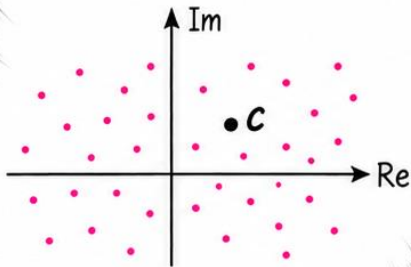
$$Z_{n+1} = Z_n^2 + C$$

Fractals from a Simple Rule

A simple rule can create infinite, multi-scale patterns.

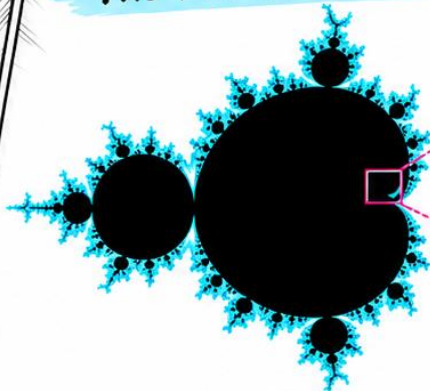
Simple iteration rule

$$z_{n+1} = z_n^2 + c$$



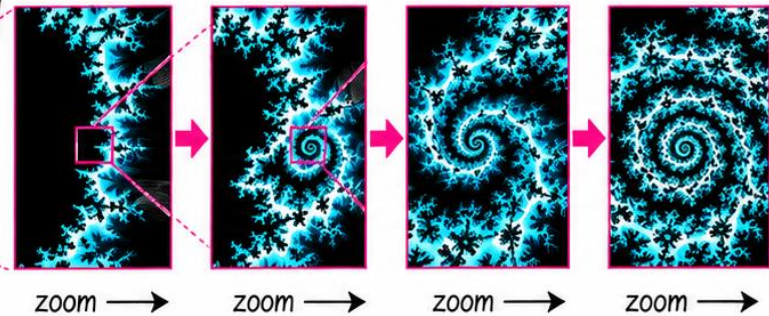
Each point c in the complex plane generates a sequence.

The Mandelbrot set



The set of points c for which the sequence remains bounded forms a fractal boundary.

Multi-scale self-similarity



The same pattern appears at different scales.

Fractals in nature



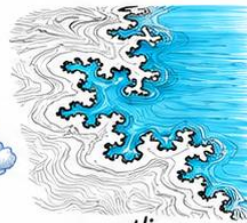
fern



tree



cloud



coastline

Fractals describe irregular shapes with structure across scales.

Question:
What is the dimension of the above data?

“How Long Is the Coast of Britain? Statistical Self-Similarity and Fractional Dimension” by Benoit Mandelbrot

Power law is the most fundamental law in data science – **why?**

Fractal dimension: a power law for geometric complexity

Fractal dimension measures how the amount of structure changes with scale.

$$N(s) \propto s^{-D}$$

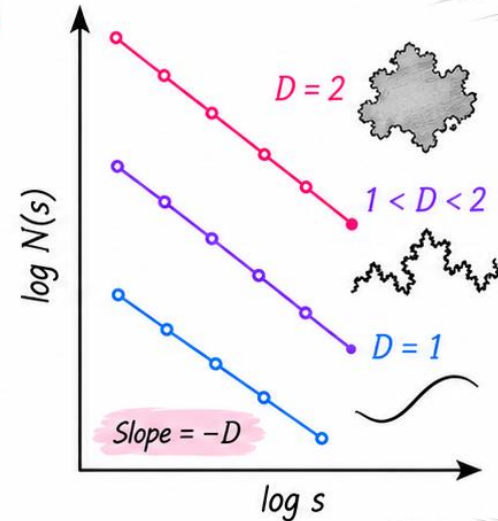
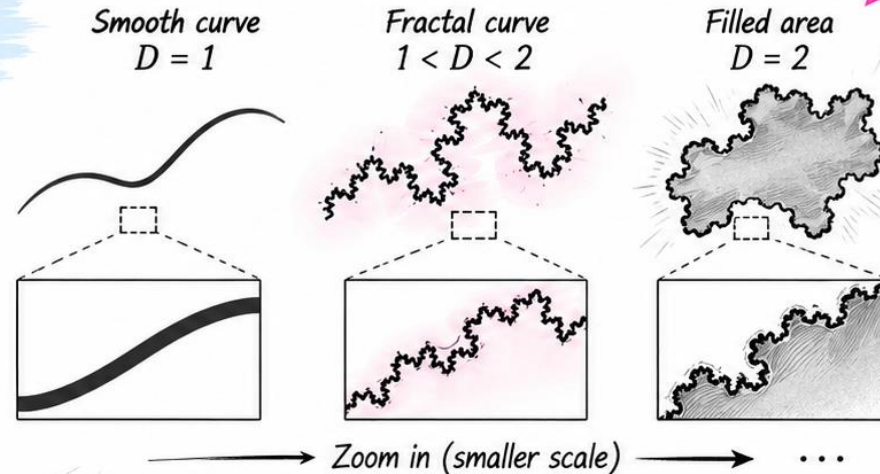
$N(s)$: amount of structure needed to describe the object at scale s

D : fractal dimension

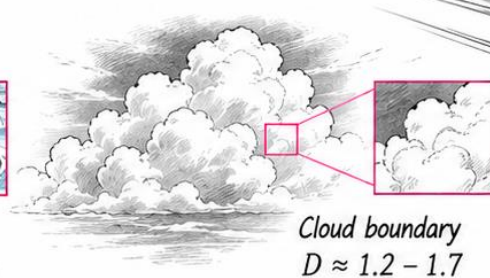
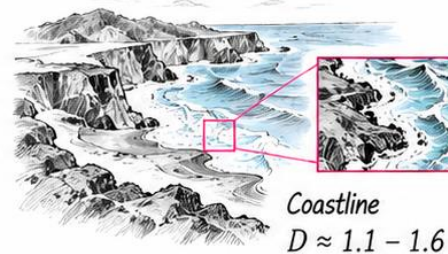
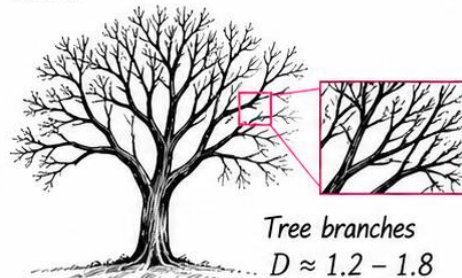
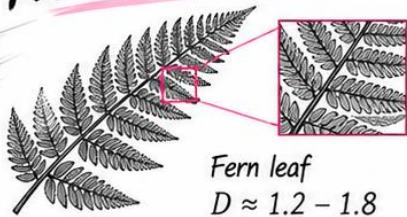
Equivalently,

$$D = - \lim_{s \rightarrow 0} \frac{\log N(s)}{\log s}$$

Increasing geometric complexity



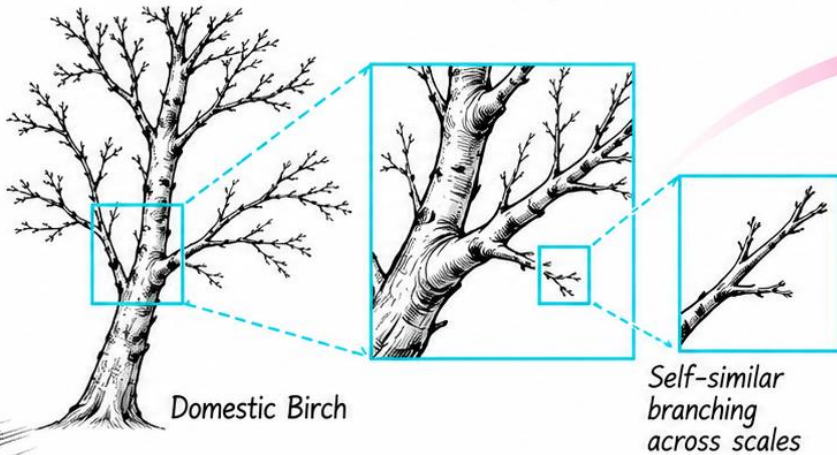
Fractals in nature



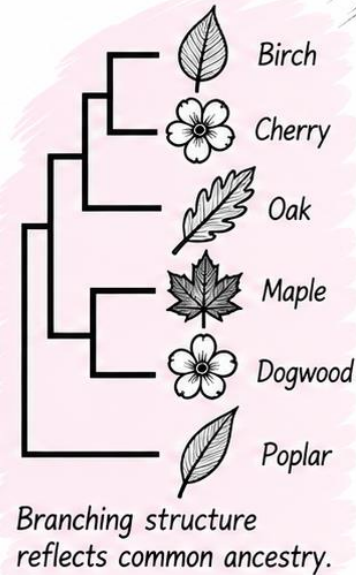
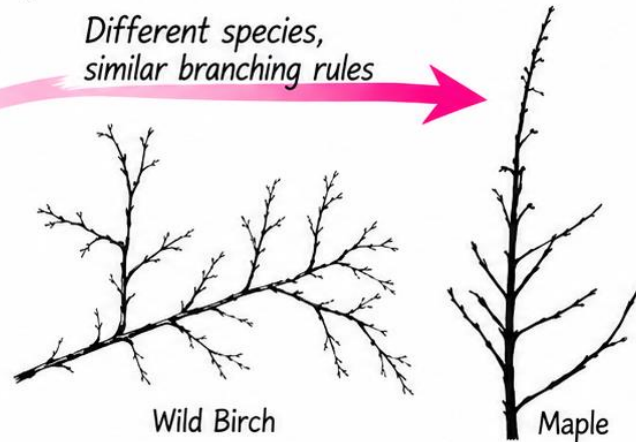
John Napier (1550–1617), the inventor of logarithms
 λόγος (logos) meaning proportion and ἀριθμός (arithmos) meaning number

Fractal dimension and evolution

Tree branching patterns exhibit fractal geometry that reflects evolutionary growth.



Different species,
similar branching rules



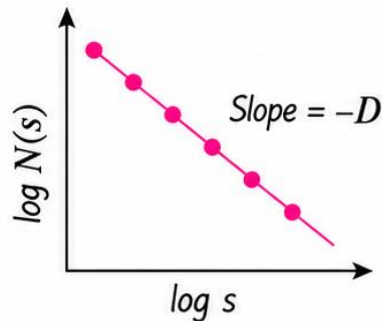
Fractal dimension

Branches are fractal-like: complexity increases as we look at finer scales.

$$N(s) \propto s^{-D}$$

amount of structure at scale s

fractal dimension



Fractal dimension of tree branching (example)

Type	Domestic	Wild
Birch	1.262	1.113
Cherry	1.647	1.459
Oak	1.648	1.279
Maple	1.378	1.358
Dogwood	1.656	1.456
Poplar	1.424	1.456

Fractal dimension quantifies branching complexity and can be used to compare species.

Box Counting Method

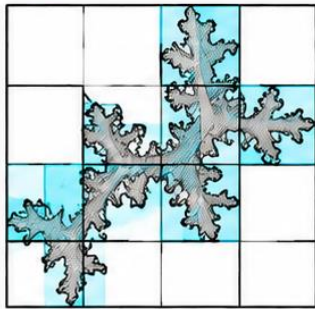
Cover the object with a grid of boxes and count how many boxes contain the object.

$$N(s) \propto s^{-D}$$

$N(s)$: number of boxes that contain the object at box size s

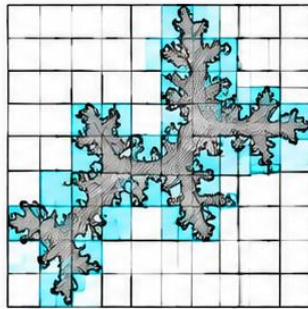
D : fractal dimension

Large boxes



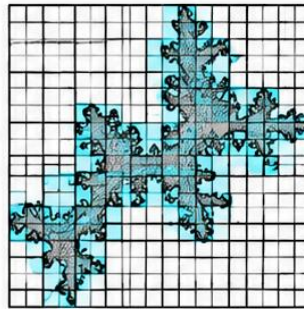
s $N(s) = 6$

Smaller boxes



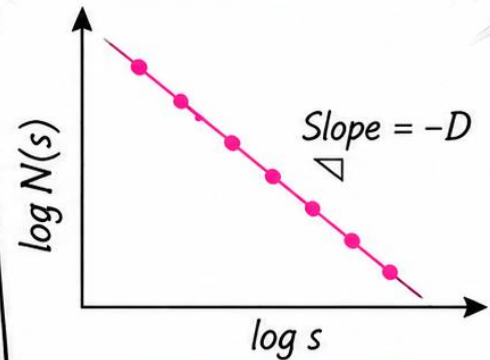
$s/2$ $N(s/2) = 20$

Even smaller boxes



$s/4$ $N(s/4) = 68$

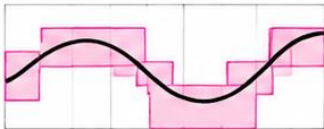
Log-log plot



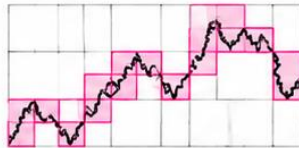
$$D = -\lim_{s \rightarrow 0} \frac{\log N(s)}{\log s}$$

Examples

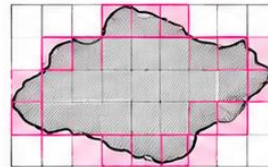
Smooth curve ($D = 1$)



Fractal curve ($1 < D < 2$)



Filled region ($D = 2$)



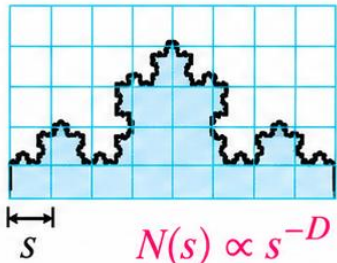
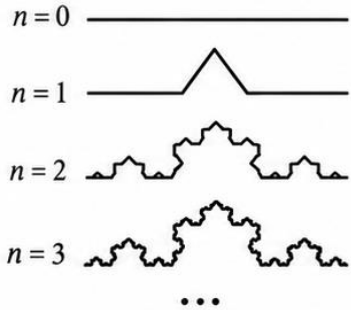
Key idea

As the box size s decreases, the number of required boxes increases following a power law. The slope on a log-log plot gives the fractal dimension.

Box Counting Method: Examples

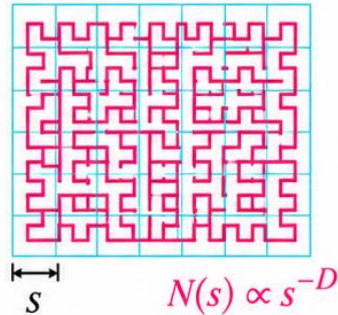
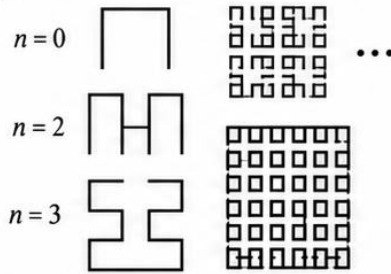
Count how the number of boxes scales with the box size: $N(s) \propto s^{-D}$

Koch curve



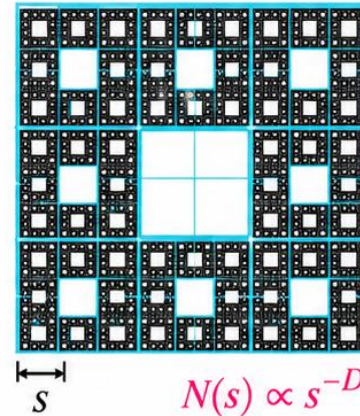
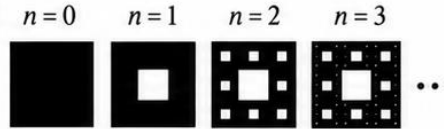
$$D = \frac{\log 4}{\log 3} \approx 1.2619$$

Hilbert curve



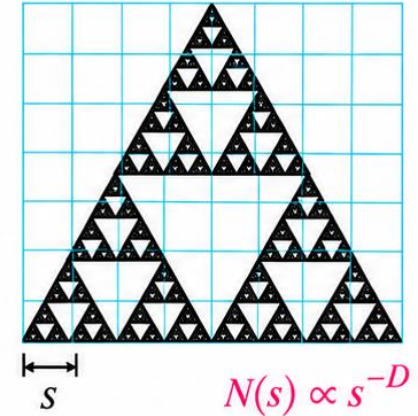
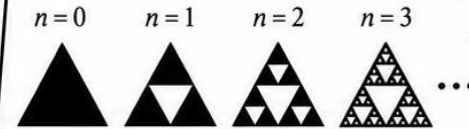
$$D = \frac{\log 4}{\log 2} = 2$$

Sierpinski carpet



$$D = \frac{\log 8}{\log 3} \approx 1.8928$$

Sierpinski triangle



$$D = \frac{\log 3}{\log 2} \approx 1.5850$$

Perimeter-area dimension (Hausdorff)

Box-counting method (Mandelbrot) is the most widely used method.

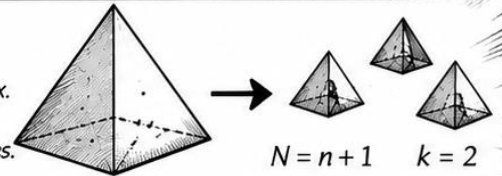
Exercise: write codes for generating these objects and compute FD

Sierpiński Simplex in n -D

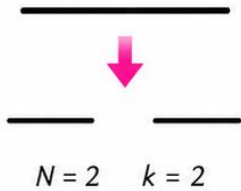
A simple recursive rule creates a fractal in any dimension.

Recursive rule

Start with an n -dimensional simplex.
Divide into 2^n smaller simplices
and keep the $(n+1)$ corner simplices.

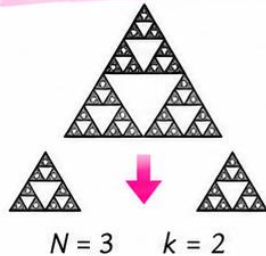


1D: line



$$D_1 = \frac{\log 2}{\log 2} = 1$$

2D: Sierpiński triangle



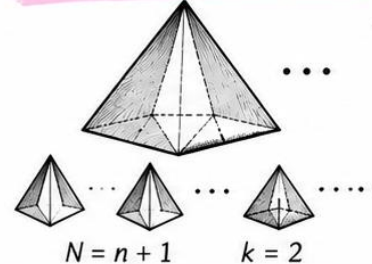
$$D_2 = \frac{\log 3}{\log 2} \approx 1.585$$

3D: Sierpiński tetrahedron



$$D_3 = \frac{\log 4}{\log 2} = 2$$

n -D: Sierpiński simplex

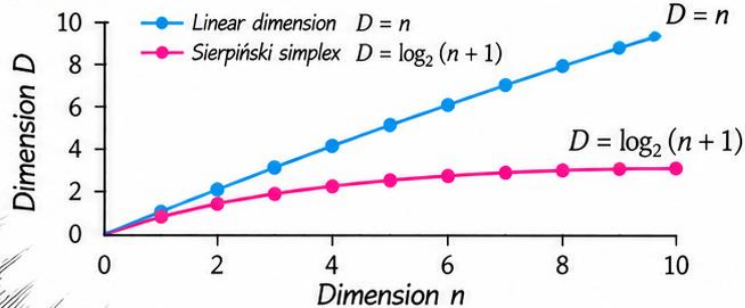


$$D_n = \frac{\log(n+1)}{\log 2} = \log_2(n+1)$$

At each step,
keep the $(n+1)$
corner simplices
scaled by $1/2$.

The same simple
rule works
in any dimension!

Fractal dimension vs. linear dimension



The fractal
dimension grows
only logarithmically,
while the ambient
dimension grows
linearly.

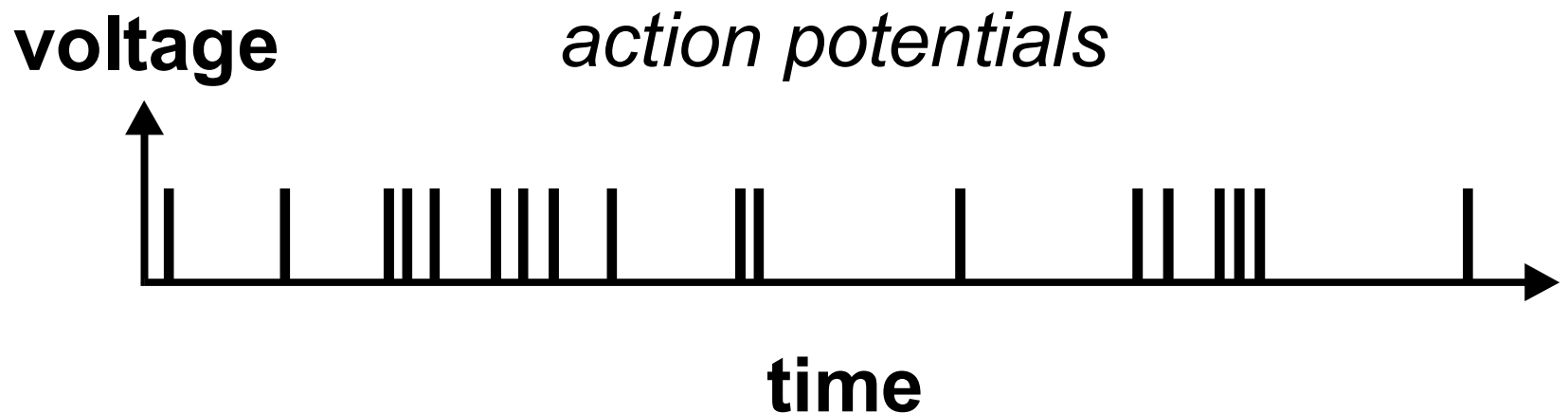
$$D_n = \log_2(n+1) \ll n \text{ (as } n \rightarrow \infty)$$

Values

n (dimension)	$N = n + 1$	$D_n = \log_2(n+1)$	Linear dimension
1	2	1.000	1
2	3	1.585	2
3	4	2.000	3
4	5	2.322	4
5	6	2.585	5
...	...	...	...

Electrical Activity of Auditory Nerve Cells

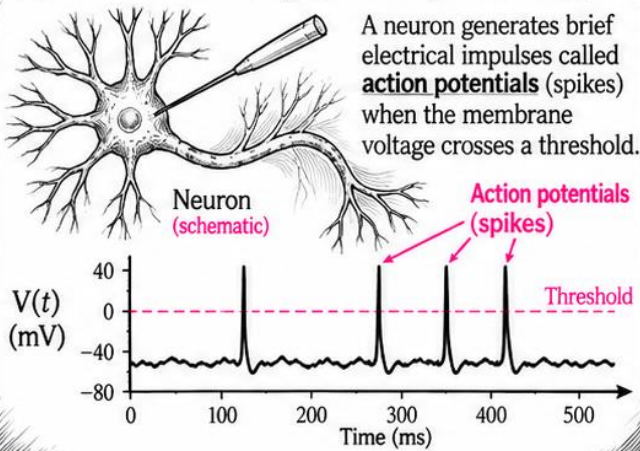
Teich, Jonson, Kumar, and Turcott 1990
Hearing Res. 46:41-52



From Neuronal Spikes to Firing Rates at Multiple Time Scales

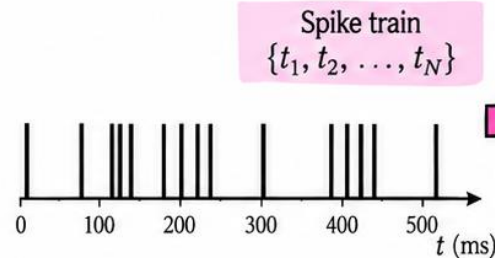
Neuronal activity can be quantified by counting spikes in time windows.

1. Neuronal firing (action potentials)



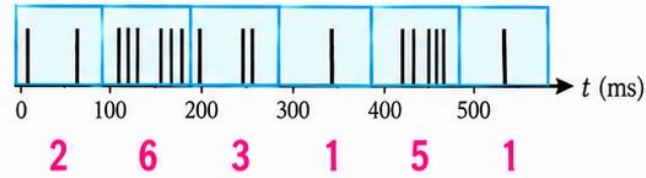
2. Spike train

Represent each spike by its time t_i .



3. Divide into time windows and count spikes

Window length Δt



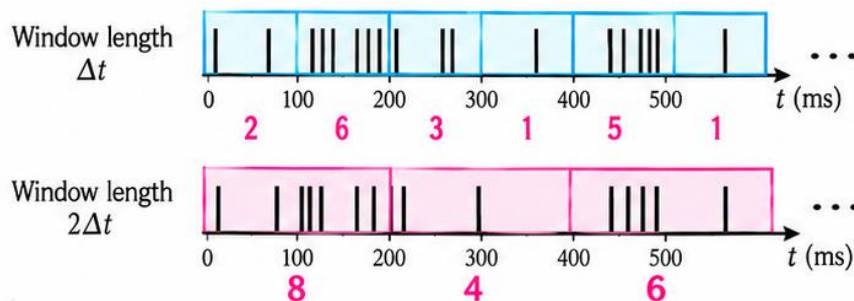
Spike counts : $n_i = 2, 6, 3, 1, 5, 1$

Firing rate : $r_i = \frac{n_i}{\Delta t}$ (spikes/s)

These are counts. Divide by Δt to get firing rates.

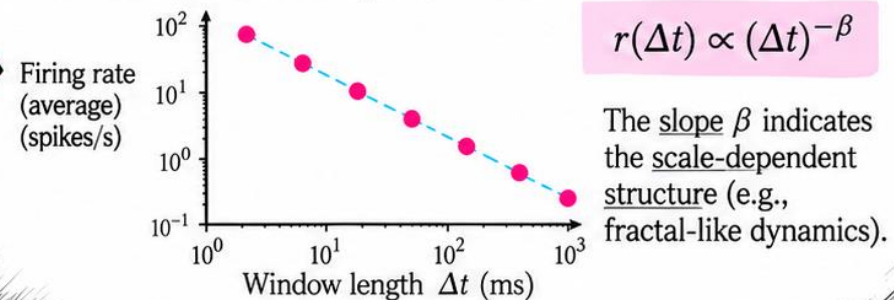
4. Repeat for different window lengths

Use larger windows (e.g., $2\Delta t$) to get a coarser firing rate.



5. Analyze the scale dependence

Compute the average firing rate for multiple window lengths and examine how it changes with scale.



Neuronal firing is a sequence of action potentials. By counting spikes in time windows of different lengths, we obtain firing rates and analyze how they change across scales.

blood vessels in the retina

Family, Masters, and Platt 1989

Physica D38:98-103

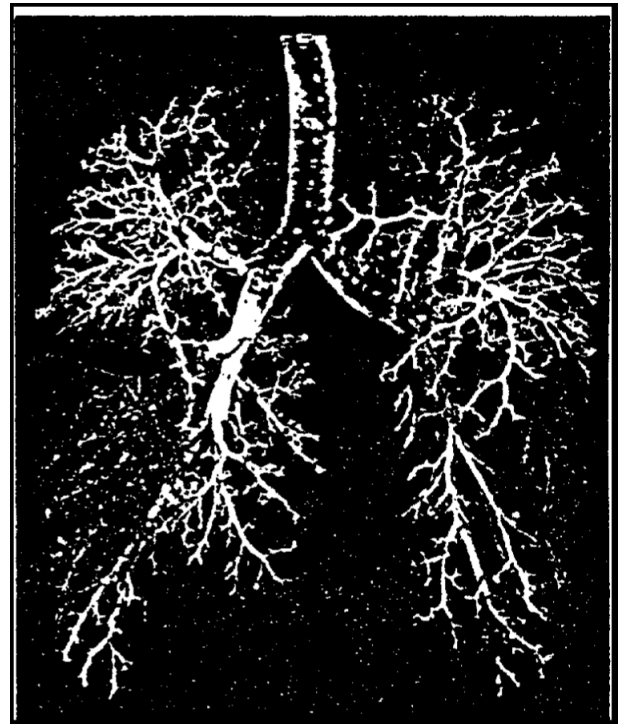
Mainster 1990 Eye 4:235-241



air ways in the lungs

West and Goldberger 1987

Am. Sci. 75:354-365

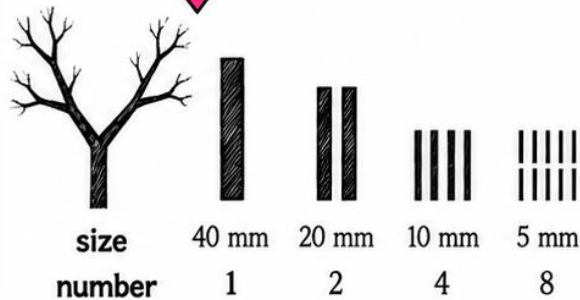
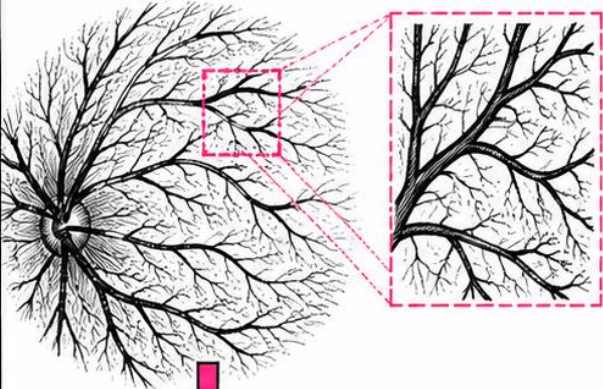


Fractal Dimension: A Probabilistic View

Same pattern across scales.

1. Example: Blood vessels in the retina

The retinal vasculature has a branching, self-similar structure.



Smaller vessels are more numerous.
The same pattern repeats at different scales.

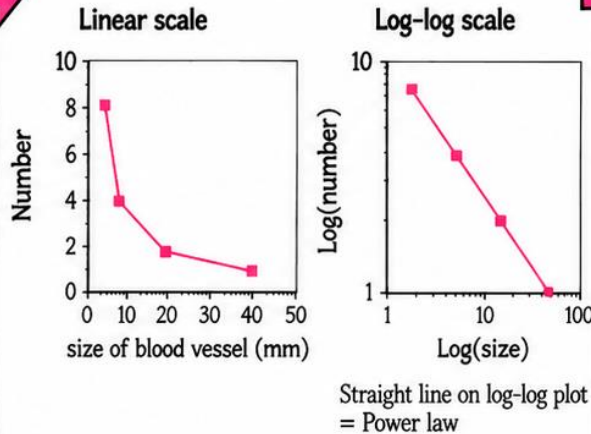
2. Power law in probability density function

The distribution of vessel sizes follows a power law.

$$p(s) \propto s^{-\alpha}$$

s : vessel size
 α : scaling exponent

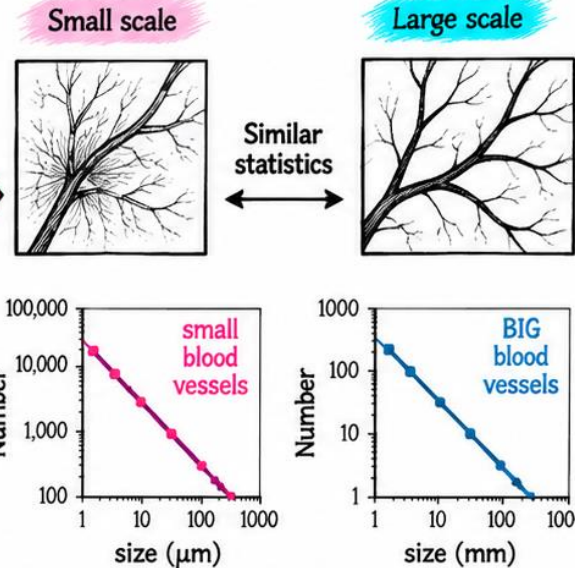
HOW OFTEN there is THIS SIZE?



A power-law distribution means there is no characteristic size. The same scaling holds across scales.

3. Probabilistic self-similarity

The statistics of the big pieces is the same as the statistics of the small pieces.



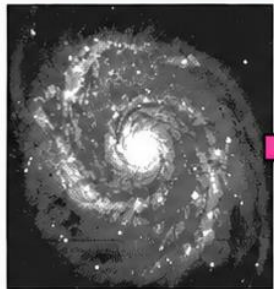
The same functional form (power law) appears at different scales.
This scale-invariance is the hallmark of a fractal, and the fractal dimension (FD) quantifies it.

Problem. Using in-class data, build trees and compute FD. Investigate the relation between the dimension of data and FD of trees.

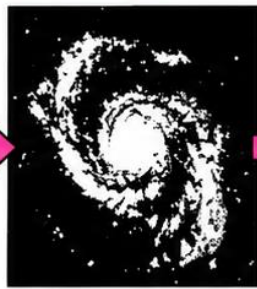
FD computation by box counting

Cover the object with boxes at multiple scales and count.

1. Preprocess the image



Gray scale image



Binary image
(threshold)



Boundary (contour)
of the object

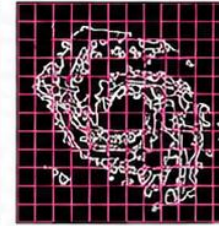
2. Box counting at different scales

Box size 3×3



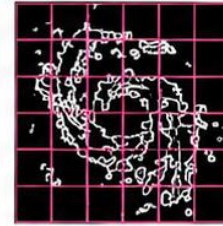
$$N(3) = 642$$

Box size 22×22



$$N(22) = 120$$

Box size 40×40



$$N(40) = 48$$

Box size 75×75



$$N(75) = 20$$

A box is counted if it contains any part of the contour (white).

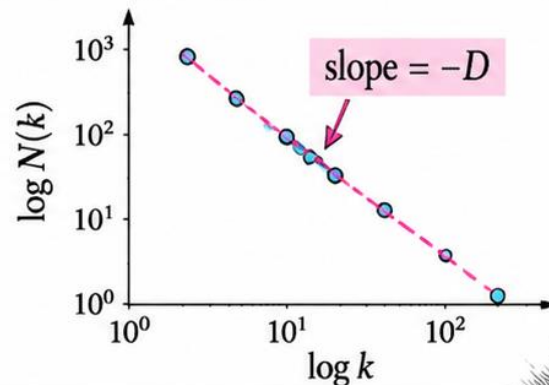
3. Count and plot

For each box size k ,
count the number of boxes $N(k)$.

$$N(k) \propto k^{-D}$$

$$D = - \frac{\text{slope of } \log N(k)}{\log k}$$

4. Log-log plot



5. Result

The fractal dimension (FD) is the negative
slope of the line in the log-log plot.

$$FD = D$$

Example (galaxy boundary): $D \approx 1.6$

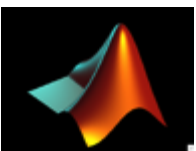
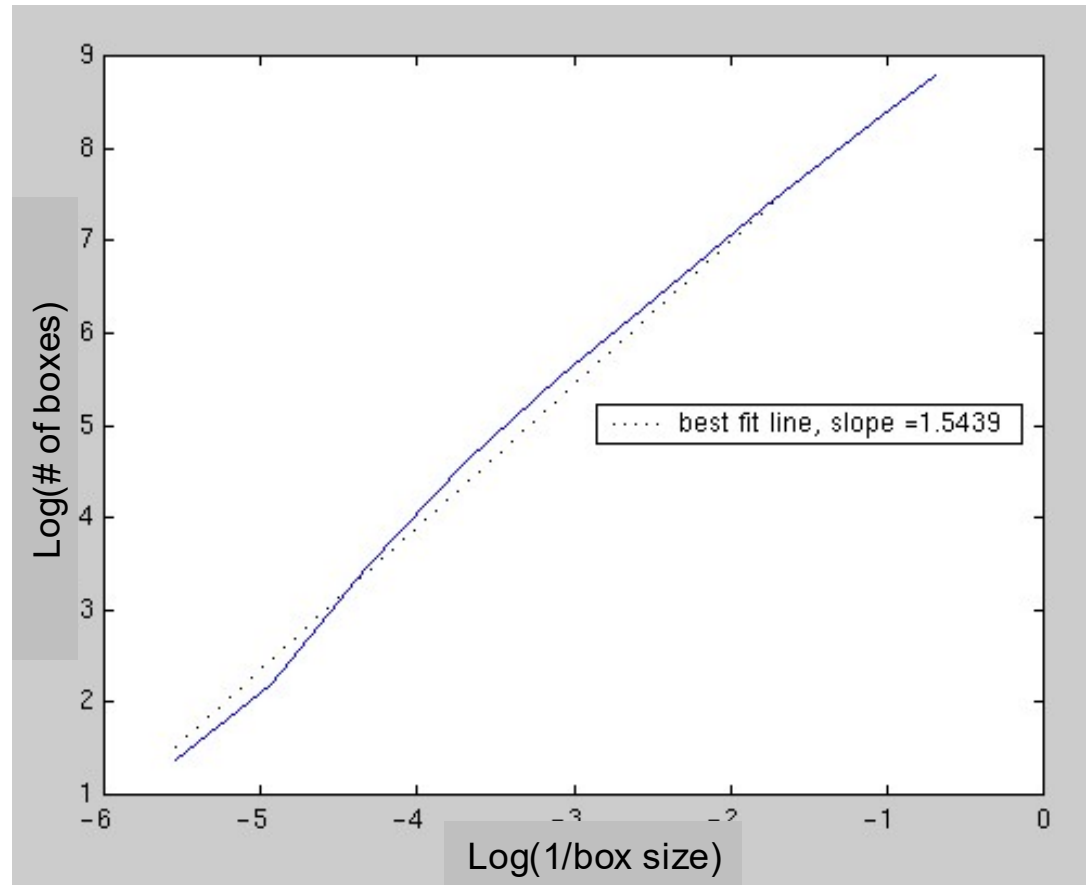
Fractal objects
have non-integer
dimensions.



Box counting measures how the number of occupied boxes scales with box size.
The fractal dimension is the (negative) slope on the log-log plot.

FD estimation via linear regression

k	N
Box size (in pixels)	number of boxes containing the contour
2	6544
3	3897
6	1562
12	591
22	250
40	101
75	32
138	9
256	4



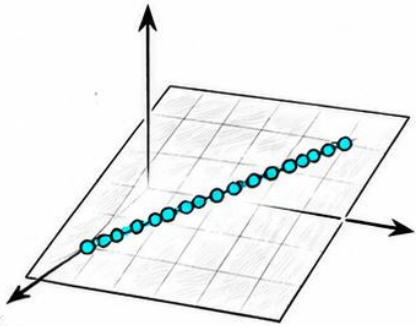
Box counting

$$FD = \text{slope} = 1.5439$$

Dimension of Point Cloud Data

How many neighbors grow with scale?

1. Intrinsic vs. embedding dimension

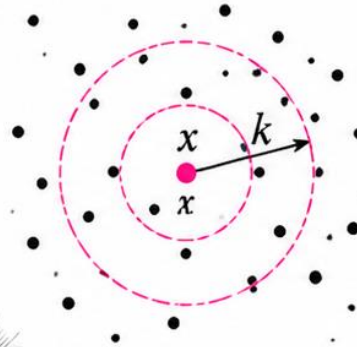


Data may lie on a low-dimensional structure (intrinsic dimension D_i) inside a higher-dimensional space (embedding dimension D_e).

$$D_i \leq D_e$$

Fractal dimension (FD) approximates the intrinsic dimension.

2. Counting neighbors at scale k



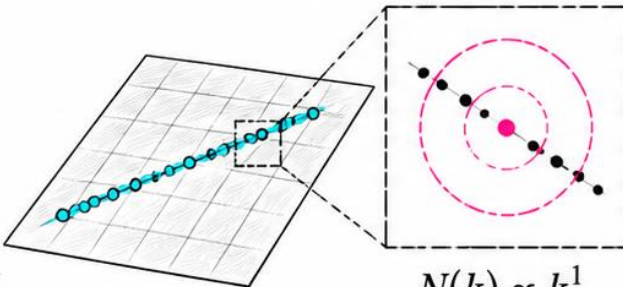
For a point x , count the number of neighbors within distance k :

$$N(k) = \text{count} \{x' : \|x - x'\| < k\}$$

If the data have (fractal) dimension D_i ,

$$N(k) \approx k^{D_i}$$

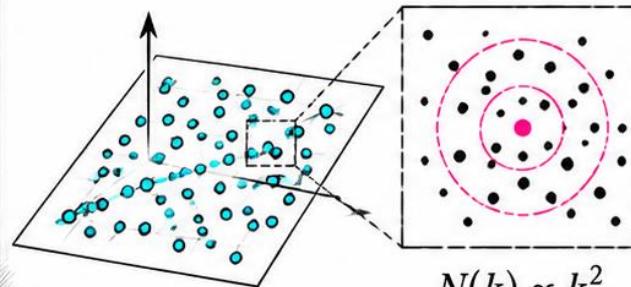
3. Example: line in a plane



- Line in a plane
- Embedding dim. $D_e = 2$
- Intrinsic dim. $D_i = 1$
- Fractal dim. $FD \approx 1$

$$N(k) \propto k^1$$

4. Example: uniform distribution in a plane



- Uniform distribution in a plane
- Embedding dim. $D_e = 2$
- Intrinsic dim. $D_i = 2$
- Fractal dim. $FD \approx 2$

$$N(k) \propto k^2$$

Fractal dimension measures how the number of neighbors grows with scale.

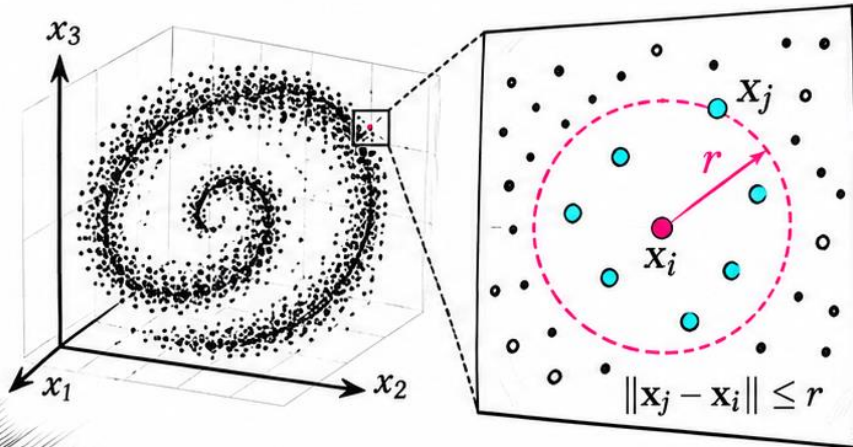
It approximates the intrinsic dimension of the data, which can be different from the embedding dimension.

Correlation Dimension

How the number of close pairs scales with distance reveals the dimensionality of the attractor.

1. Point set and pair counting

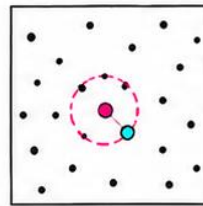
Data points $\Omega = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \subset \mathbb{R}^n$



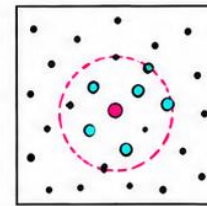
2. Correlation integral

Fraction of pairs within distance r :

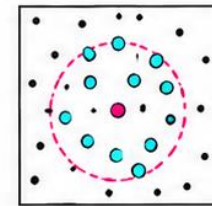
$$C_m(r) = \lim_{N \rightarrow \infty} \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=i+1}^N I(\|\mathbf{x}_j - \mathbf{x}_i\| \leq r)$$



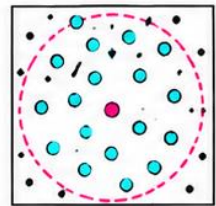
r (small)



r



larger r



even larger r

3. Scaling behavior

For many fractal sets,

$$C_m(r) \propto r^D \quad (\text{small } r)$$

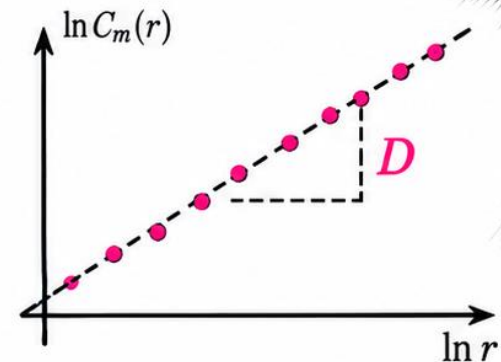
Taking logarithm,

$$\ln C_m(r) \approx D \ln r + \text{constant}$$

4. Correlation dimension

Slope of the log-log plot gives the dimension:

$$D = \lim_{r \rightarrow 0} \frac{\ln(C_m(r))}{\ln(r)}$$



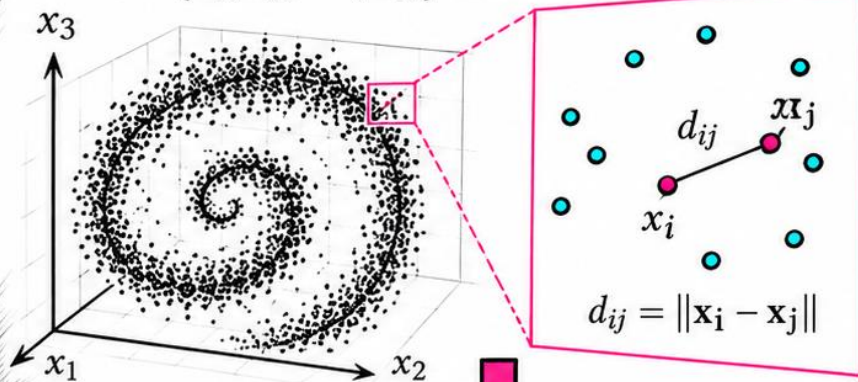
Correlation
dimension

Correlation Dimension and Persistent Homology

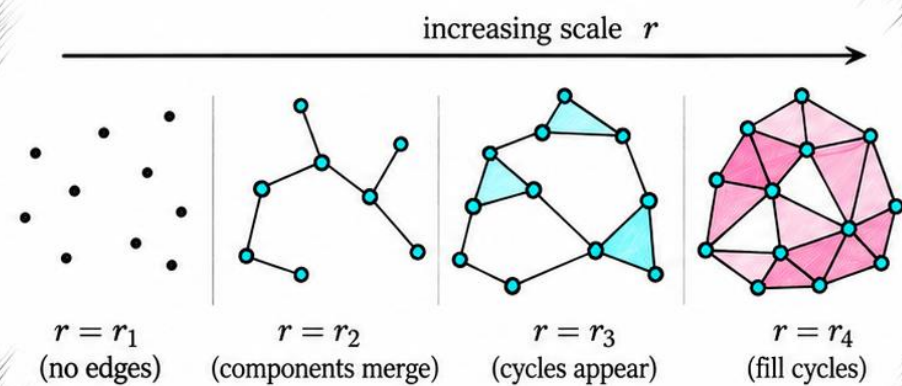
Same distance information.
Different questions.

1. Point cloud and pairwise distances

$$X = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \subset \mathbb{R}^n$$



2. Rips filtration: same distances for both methods



3. Correlation dimension: edge density scaling

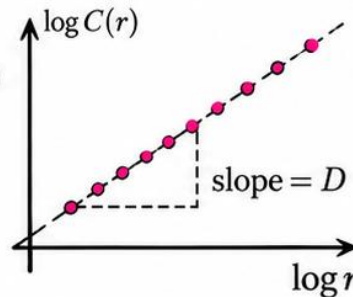
Correlation integral (fraction of pairs within r):

$$C(r) = \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=i+1}^N I(\|\mathbf{x}_j - \mathbf{x}_i\| \leq r)$$

$$C(r) = \frac{\#\{\text{edges in VR}(r)\}}{\binom{N}{2}}$$

For small r :

$$C(r) \sim r^D \Rightarrow D = \lim_{r \rightarrow 0} \frac{\log C(r)}{\log r}$$



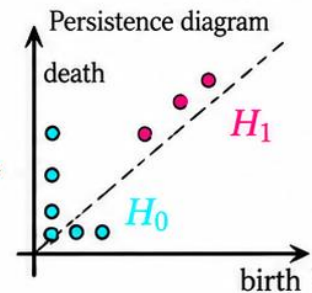
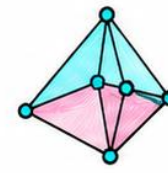
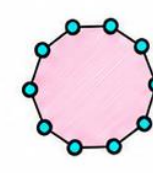
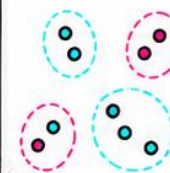
4. Persistent homology: topological organization

From the same filtration $\text{VR}(r)$, compute Betti numbers $\beta_k(r)$.

β_0 : components

β_1 : cycles (holes)

β_2 : cavities



Connection: correlation dimension uses the number of edges (1-skeleton) in the Rips filtration, while persistent homology uses the full simplicial structure to capture higher-dimensional topology.

Same distances.
Complementary information.

Problems. How correlation dimension and persistent homology are related?
Can we compute fractal dimension on a collection of cycles?

Fractal vs. Non-Fractal Data

Fractal data show scale-invariant fluctuations and a power-law behavior.

1. Non-fractal vs. Fractal

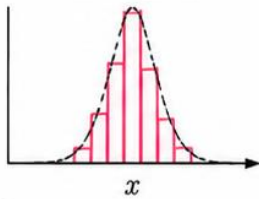
Different structures → different statistics

Non-fractal

data



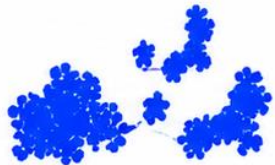
distribution



Random, characteristic scale
(e.g., Gaussian)

Fractal

data



$\log p(x)$

power law

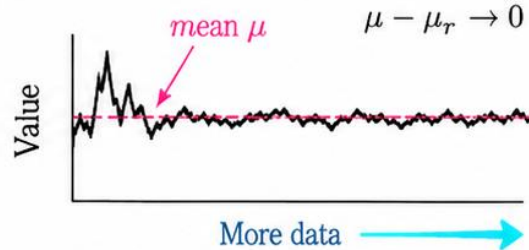
$\log x$

No characteristic scale
(scale-invariant)

2. Mean - Fluctuation behavior

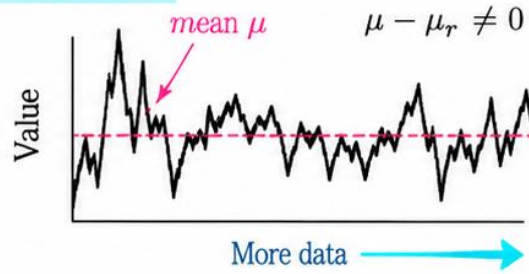
Fluctuations decrease with scale.

Non-fractal



Averaging over larger scales converges to a constant (mean).

Fractal



Fluctuations persist across scales
(no characteristic scale).

3. How to detect fractality

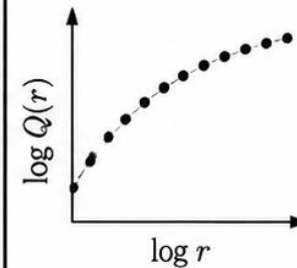
Use log-log plots to check for a power law.

Many quantities follow a power law:

$$Q(r) \propto r^{-\alpha}$$

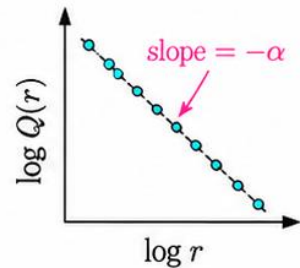
r : scale
 α : scaling exponent

Non-fractal (e.g., exponential)



Not a straight line
(has a characteristic scale).

Fractal (power law)



Straight line on log-log plot
(scale-invariant).

**A straight line in a log-log plot
indicates fractal behavior.**

Fractal data have no characteristic scale and show the same statistical pattern at all scales.

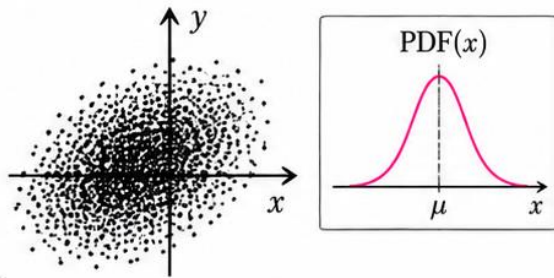
Problem. Can a non-fractal dataset exhibit fractal structure over a range of scales?

Can non-fractal data show fractal behavior over a restricted range?

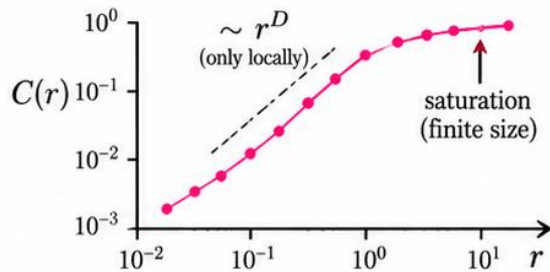
Yes.
Local scaling.

1. Global structure: non-fractal dataset

The whole dataset has a characteristic scale (e.g., Gaussian).



Correlation integral



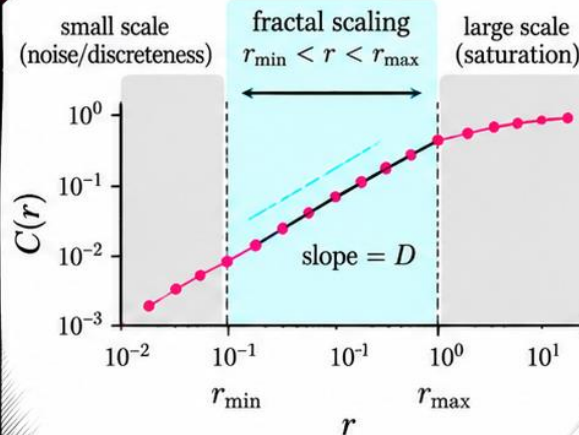
No power-law over all scales
→ globally non-fractal.

2. Local or intermediate-scale fractal behavior

Over a restricted range of scales $r_{\min} < r < r_{\max}$, the data can show approximate power-law scaling.

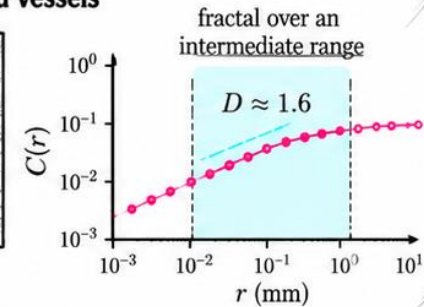
$$C(r) \propto r^D \quad (r_{\min} < r < r_{\max})$$

On a log-log plot, this appears as a straight line only in this range.

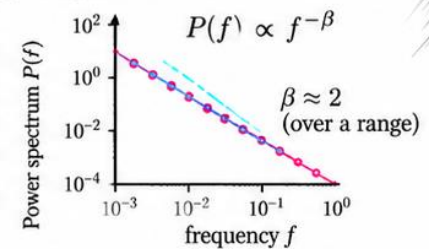
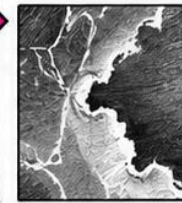


3. Real-world examples

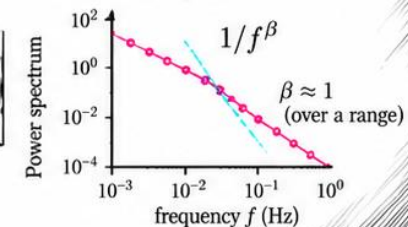
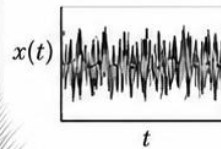
(a) Retinal blood vessels



(b) Natural image (e.g., coastline, clouds)



(c) Brain fMRI time series (example)



A dataset can be globally non-fractal but still exhibit fractal behavior over a restricted subset or range of scales.

Fractality is about scale-dependent patterns, not an all-or-none property.

Ordinary Coin Toss vs. St. Petersburg Game

Same random mechanism but very different statistics.

1. Ordinary coin toss

Simple, well-behaved distribution



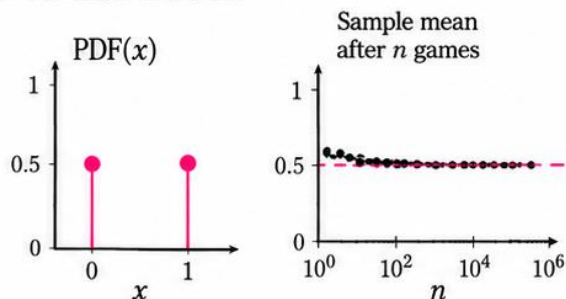
Toss a coin.

Head → win \$1

Tail → win \$0

Average (expected value)

$$E[X] = 0.5$$



Sample mean → 1/2

Non-fractal (has a characteristic scale)

2. St. Petersburg Game (Niklaus Bernoulli)

Rare large rewards → heavy-tailed distribution



Toss a coin until the first head appears.
If the head appears on the N -th toss,
win $\$2^N$.

H	→	\$2
TH	→	\$4
TTH	→	\$8
TTTH	→	\$16
⋮		⋮

$$P(N) = 2^{-N}, \quad X = 2^N$$

$$\begin{aligned} E[X] &= \sum_{N=1}^{\infty} 2^N 2^{-N} \\ &= \sum_{N=1}^{\infty} 1 \\ &= \infty \end{aligned}$$

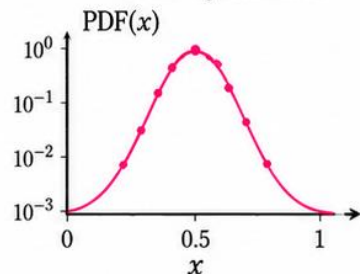
Infinite average!

Fractal / heavy-tailed

3. Distributions

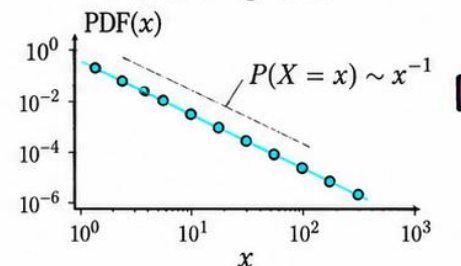
Ordinary coin toss has a thin-tailed distribution.

Ordinary coin toss



Finite variance, characteristic scale.

St. Petersburg Game



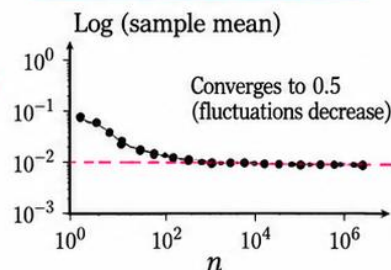
Power-law (heavy tail), no characteristic scale.

Fractal-like

4. Scaling of the sample mean

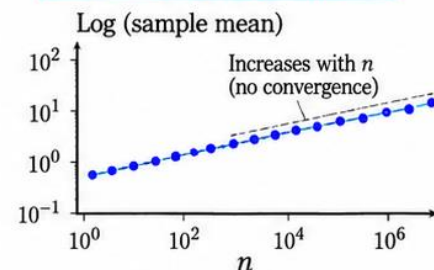
How the average behaves as we collect more data.

Ordinary coin toss



Non-fractal behavior

St. Petersburg Game



Fractal / heavy-tailed behavior

Same random process (coin toss) can lead to very different statistics.

Problem. St. Petersburg game has FD of 1. Design a coin tossing game that has FD close to 2. Is it even possible? This relates to **algorithmic complexity**.

From St. Petersburg to Higher-Dimensional Games

Can we design a coin game with FD close to 2?

1. St. Petersburg Game (FD ≈ 1)

One-dimensional heavy tail

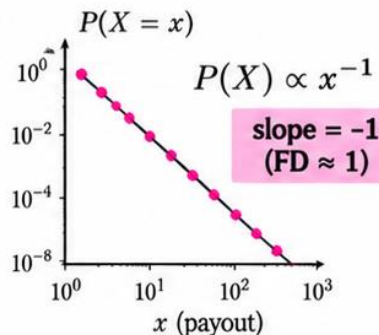


Toss a fair coin until the first head.

If head appears on the N -th toss, win 2^N .

$$P(N = n) = 2^{-n}$$

$$X = 2^N$$



- Power-law with exponent 1
- Infinite mean
- Fractal dimension (of distribution) ≈ 1

2. A Two-Dimensional Coin Game (FD ≈ 2)

Use two independent geometric counts

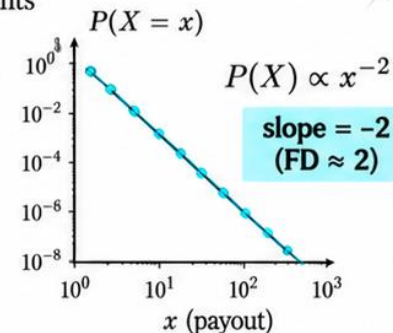
Stream A $\dots \rightarrow \frac{H}{N_1}$

Stream B $\dots \rightarrow \frac{H}{N_2}$

Toss two independent fair coins until each gets a head.
Win $X = 2^{N_1 + N_2}$.

$$P(N_1 = n_1, N_2 = n_2) = 2^{-(n_1 + n_2)}$$

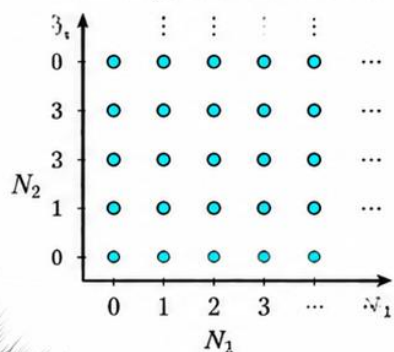
$$P(X = x) \propto x^{-2}$$



- Power-law with exponent 2
- Still infinite mean (marginal $\propto 1/x^2$)
- Generalize to k streams $\rightarrow$ exponent k (FD $\approx k$)

3. Geometric view and fractal dimension

Set of possible outcomes in log-scale



Each point $(\bar{n}_1, \bar{n}_2)$ has probability $2^{-(n_1 + n_2)}$.

The number of outcomes with $N_1 + N_2 \leq n$ is

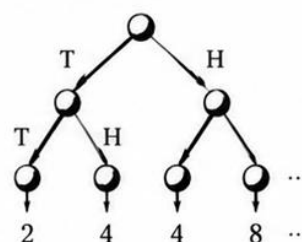
$$\# \{n_1 + n_2 \leq n\} \approx n^2.$$

$$N(r) \propto r^2 \rightarrow \text{FD} = 2$$

In k dimensions, $N(r) \propto r^k \rightarrow \text{FD} = k$.

4. Relation to algorithmic complexity

Rarer large outcomes require longer descriptions.



To specify an outcome with payout $X = 2^n$, we need to describe a string of length n (T...TH).

Kolmogorov (algorithmic) complexity

$K(\underline{X})$ = length of the shortest program to output $\underline{X}$.

$$K(2^n) \approx n \approx \log_2 X$$

For $P(X) \propto X^{-\alpha}$, we have

$$P(X) \approx 2^{-\alpha K(X)}.$$

$\alpha = 1 \rightarrow$ St. Petersburg (FD ≈ 1)

$\alpha = 2 \rightarrow$ Two-stream game (FD ≈ 2)

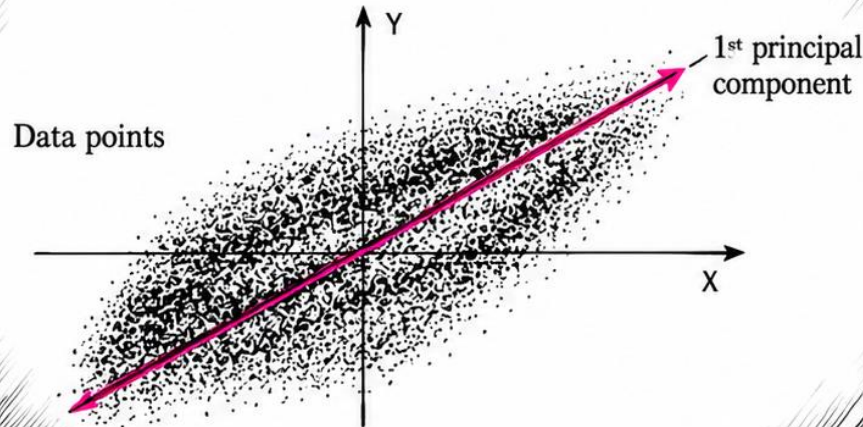
We can design coin tossing games with FD ≈ 2 (or k) using multiple independent streams.

This links heavy-tailed distributions, fractal dimension, and algorithmic complexity.

FD, PCA and dimensionality

Different notions of dimension.

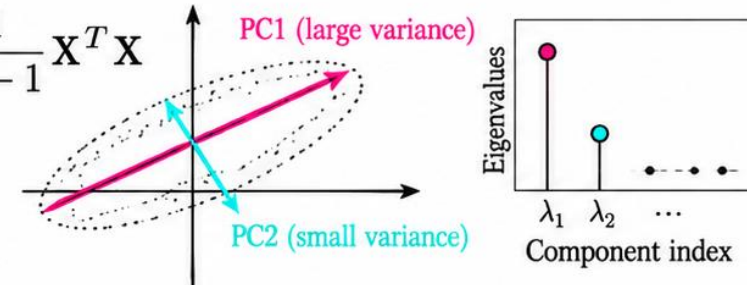
1. Data points in 2D



2. Principal component analysis (PCA)

Eigen-decomposition of the covariance matrix

$$\Sigma = \frac{1}{N-1} \mathbf{X}^T \mathbf{X}$$

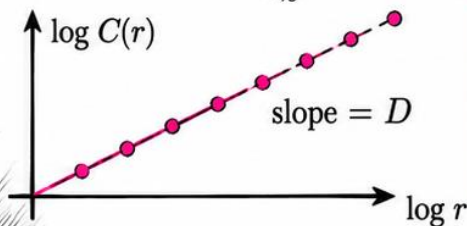


Number of significant principal components = number of large eigenvalues (e.g., via scree plot or variance threshold).

3. Fractal dimension (FD)

How the number of points within radius r scales:

$$C(r) = \frac{1}{N^2} \sum_{i,j} I(\|\mathbf{x}_i - \mathbf{x}_j\| \leq r) \sim r^D$$



FD measures how the data fills space at multiple scales. For an elongated 2D cloud, $D \approx 2$.

4. Relation and interpretation

PCA dimensionality

- Global, linear measure
- Number of significant PCs = number of large eigenvalues
- For the shown data: 1 significant PC (points lie near a line)

PCA captures linear structure (variance along directions).

Fractal dimension

- Scale-dependent, geometric measure
- Describes how densely the points fill space
- For the shown data: $D \approx 2$ (points fill a 2D region, not a 1D curve)

FD captures intrinsic geometric dimensionality (nonlinear).

They measure different things: PCA = number of linear directions, FD = how the set fills space across scales.

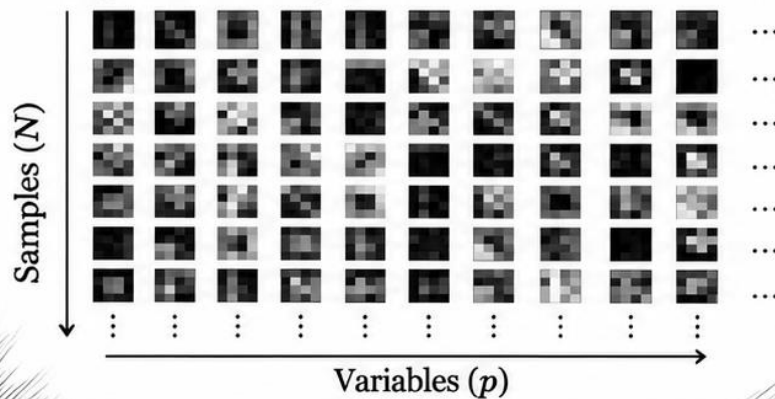
Problem. How FD and the number of significant principal components are related?

Factor Analysis, FD and Dimensionality

High-dimensional data can be explained by a few latent factors.

1. High-dimensional data

N samples with p correlated variables (e.g., flattened images)



2. Factor analysis model

Observed variables are generated from a few latent factors.

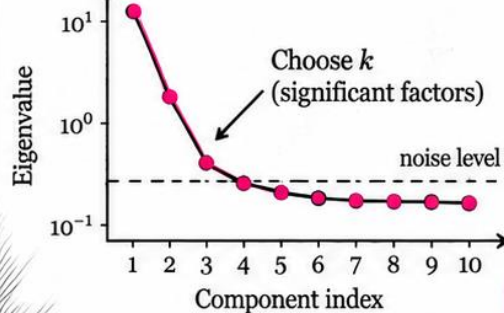
$$\mathbf{X} = \mathbf{\Lambda} + \mathbf{F} + \mathbf{E}$$

observed data ($N \times p$) factor loadings ($p \times k$) latent factors ($N \times k$) noise ($N \times p$)

The diagram shows the equation $\mathbf{X} = \mathbf{\Lambda} + \mathbf{F} + \mathbf{E}$ with corresponding matrix dimensions. Below the equation, a stack of images represents $\mathbf{X}$ ($N \times p$). To its left, two small square matrices represent $\mathbf{\Lambda}$ ($p \times k$). To its right, a matrix with vertical stripes represents $\mathbf{F}$ ($N \times k$). Further right, a matrix with horizontal stripes represents $\mathbf{\Lambda}^T$ ($k \times p$). An approximation symbol $\approx$ is placed between $\mathbf{F}$ and $\mathbf{\Lambda}^T$.

3. Determining the number of significant factors

Eigenvalues of the covariance matrix



Other criteria

- Scree plot (elbow)
- Variance explained (e.g., 80–90%)
- Parallel analysis (compare with random data)
- Information criteria (AIC, BIC)

4. Factor dimension vs. fractal dimension

Factor dimension (FA)

- Global linear structure
- Number of significant factors k
- Equivalent to intrinsic dimension of a low-dimensional linear model
- For the shown data: k is small (e.g., 2–4)

Captures linear correlations between variables.

Fractal dimension (FD)

- Scale-dependent geometric measure
- Based on pairwise distances $C(r) \sim r^D$
- Describes how the data fills space (possibly nonlinear)
- For the shown data: D can be non-integer and larger than k (e.g., $D \approx 4-8$)

Captures overall geometrical complexity (including nonlinear structure).

FA and FD answer different questions:

FA: how many latent factors explain the correlations?

FD: how the data fills space across scales?

Problem. How FD and factor dimension are related?

Data Complexity

Unit outcome

1. Understand fractal dimension
2. Understand how to relate data scale and data dimension